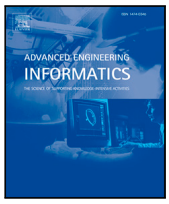
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Full length article



Bearing fault diagnosis method based on multi-level information fusion ☆

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ABSTRACT

Bearing fault diagnosis is crucial for saving time and reducing costs, as it targets one of the most critical components in rotary machines. The decision fusion method has proven to be an effective approach to improving the performance of fault diagnosis. However, the conflicting results will affect the accuracy of the final classification. This paper proposes a multi-level information fusion network (MLIFN) to improve the classification performance of multiple classifiers in decision fusion through feature fusion method. A two level decision fusion of deep features and empirical features are applied to improve the classification accuracy. Furthermore, an innovative hybrid attention mechanism, combining self-attention and mutual attention, is introduced for deep feature extraction. The Paderborn bearing dataset is employed to validate the effectiveness of the MLIFN. The results demonstrate that the diagnostic accuracy of the MLIFN can reach up to 99.82%, significantly outperforming other methods.

1. Introduction

As the industry continues to evolve, the significance of bearings in mechanical equipment has escalated. Complex industrial equipment operates in severe environments, leading to a range of bearing faults that can result in substantial maintenance expenses and potential casualties [1]. According to statistics, 40% of faults in large systems and a staggering 90% in smaller systems stem from bearing defects [2]. Consequently, the timely and precise diagnosis and resolution of bearing faults are paramount to prevent heavy losses in the industry.

Existing research in fault diagnosis primarily falls into two categories: model-based methods and data-driven methods [3]. However, due to the challenging operating conditions of complex industrial equipment and the significant reliance on specialized knowledge, establishing precise models for rotating machinery can be a daunting task. Unlike model-based approaches, data-driven fault diagnosis methods operate in an end-to-end fashion, exhibiting superior performance in intricate rotating machinery systems [4]. Furthermore, with the ongoing advancement of intelligent manufacturing, acquiring operational data from mechanical equipment has become increasingly easier and more efficient, presenting a significant opportunity for data-driven fault diagnosis [5].

Machine learning (ML) emerges as a prominent approach in data-driven fault diagnosis, exhibiting impressive performance in diagnosing faults. Wang et al. proposed a fault diagnosis technique for rolling bearing based on Bayesian optimized support vector machine (SVM), which showed good performance on fault mode diagnosis as well as fault degree distinction [6]. Tang et al. proposed a k-nearest neighbour (k-NN) of supervised learning method for fault diagnosis which achieved an accuracy of 98.9% [7]. For fault diagnosis of rolling bearing, Wang et al. proposed a hybrid approach of a random forest (RF) classifier, and the effectiveness of the method was verified [8]. applied extreme learning machine (ELM) optimized by particle swarm optimization (PSO) to bearing fault diagnosis, which could extract more prominent features than the other methods [9]. However, the substantial time investment and specialized knowledge required for feature design in these conventional ML methods present a limitation to their widespread application [4].

As ML methods continue to advance, deep learning (DL) methods have been proposed to address the aforementioned limitation. Ruan et al. proposed a convolutional neural network (CNN) design strategy based on the physical characteristics of bearing acceleration signals to optimize the parameters of CNN for better performance of bearing fault diagnosis [10]. To deal with the noise of industrial environment in the data, Chen et al. proposed a stacked de-noising auto-encoder (AE) based approach for bearing fault diagnosis [11]. In order to solve the challenge of parameter determination, Xu et al. employed sparrow search

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algorithm (SSA) to optimize the structural and training parameters of deep belief network (DBN) [12]. You et al. fused quadratic neural network and long short-term memory network (LSTM) for bearing fault diagnosis to improve the efficiency and interpretability [13]. To take full advantage of different deep learning methods, they are combined for fault diagnosis to fully exploit their potential. Yan et al. integrated the convolutional block attention module to variational autoencoder, which could improve the accuracy and anti-noise robustness of fault diagnosis [14]. Besides, DL methods are applied for health monitoring in various industrial applications. For example, Zhao et al. proposed a data-driven time–frequency technique for wind turbines fault diagnosis under time-varying speeds, and showed high time–frequency resolution in characterizing nonstationary signals [15]. Cai et al. proposed a multi-scale dynamic graph mutual information network for health monitoring of planet bearing, and showed ideal diagnostic accuracy under imbalanced data [16]. However in reality production, the complex structure of the equipment and harsh operating conditions will lead to its strong coupling, making it difficult to obtain complete and accurate information from the single sensor signal for fault diagnosis [17].

Data fusion technique has been proved to be able to improve the performance of fault diagnosis significantly, which includes data-level, feature-level and decision-level fusion. In data-level fusion, Xie et al. applied a conversion method to convert multiple signals to RGB image for fusion of multi-signal data [4]. To tackle the challenge of neglecting the interdependence between data samples, Li et al. proposed a correlation variance contribution method to conduct data-level fusion for rolling bearing fault diagnosis [18]. In feature-level fusion, Yang et al. proposed a feature fusion GAN to generate time–frequency image, which had ability to improve the diagnostic accuracy under limited samples significantly [19]. To solve the problem of insufficient feature extraction and utilization, Shu et al. proposed a feature fusion residual CNN with a double branch for fault diagnosis of broken rotor bar in induction motor [20]. In decision-level fusion, Tong et al. utilized the fuzzy rank to conduct the decision fusion with considering the confidence of multiple different sensors [21]. Xu et al. proposed a dynamically enhanced weighted voting strategy to achieve more accurate and reliable diagnostic results [22]. Data-level fusion realizes maximal preservation of the original information, but a highly consistent data structure is required to work, which is hardly guaranteed [23]. Besides, data-level fusion often does not take full advantage of the complementarity between multiple sensor data, and the raw data often contain a large amount of redundant information [21]. A major advantage of feature-level fusion is the flexibility in choosing where to fuse. But the design of feature extractor needs to be carefully considered, otherwise it will lead to further accumulation of errors [21]. Decision-level fusion can solve this problems, because the errors of the fused models come from different base learning models, and these errors are often uncorrelated. But the conflicting results may affect the accuracy of the final classification [23]. The conflicting results is caused by the poor classification performance of multiple classifiers.

To solve this issue, a multi-level information fusion network (MLIFN) is proposed in this paper, in which feature fusion is applied to improve the classification performance of multiple classifiers. The main contribution are as follows.

(1) A novel multi-level information fusion framework is proposed. The empirical features are creatively applied to deep learning networks for better performance of fault diagnosis.

(2) A hybrid attention is proposed innovatively for deep feature extraction and fusion. The self attention weights the features through its own fault information, while the mutual attention fully consider the fault information of another sensor signal to weight the features. Through the interaction of feature extraction, the correlation among the features extracted from multi-sensor data can be mined.

(3) A fusion method based on standard deviation weighting strategy is proposed for decision fusion of deep features, which can effectively improve the effect of decision fusion when combined with the decision fusion method based on Dempster–Shafer (DS) theory.

(4) Comparative experiments are conducted with the Paderborn bearing data set. The effectiveness and superiority of the MLIFN are validated.

Section 2 introduces the basic theoretical knowledge of attention mechanism and DS theory. Section 3 explains the details of MLIFN. Section 4 conducts comparative experiments. Section 5 summarizes the paper.

2. Theoretical background

2.1. Attention mechanism

To effectively prioritize the significance of various fault features, the attention mechanism computes attention coefficients to reweight these features adaptively. Its fundamental concept is illustrated in Fig. 1.

Assume the input feature set is $F = [f_1, f_2, \dots, f_K] \in R^{m\times K}$, where $f_i \in R^{m\times 1}, i = 1, 2, \dots, K$ is the ith feature. Initially, average pooling is applied to aggregate the K features into a summary representation $f_i' \in R^{1\times 1}$, and the input feature becomes $F' = [f_1', f_2', \dots, f_K'] \in R^{1\times K}$, which is expressed as

$$f_i' = AverPool(f_i) = \frac{1}{1\times m}\sum_{j=1}^{m} x_i(j), i = 1, 2, \dots, K \tag{1}$$

Next, the aggregated features undergo a two-stage dimensional transformation using predefined operations, described as follows

$$G = T(T(F') \in R^{1\times(K/r)}) \in R^{1\times K} \tag{2}$$

where $T(\cdot)$ is transformation functions, and the reduction ratio r defines the compression level. These transformations adjust the dimensionality of the input feature, yielding an attention coefficient $G = [g_1, g_2, \dots, g_K]$ that captures essential information.

Finally, the original features F are reweighted with the computed attention coefficients

$$F = F \cdot G = [f_1 g_1, f_2 g_2, \dots, f_K g_K] \tag{3}$$

This process enables the attention mechanism to amplify significant components within the original features, thereby enhancing the model's ability to focus on critical information.

2.2. Dempster-Shafer theory

DS theory is an effective uncertainty reasoning method, and is also known as evidence theory, which fuse inaccurate and incomplete information to obtain more accurate and credible results [24]. The DS theory is shown as follows: Define nonempty set $\Theta = A_1, A_2, \dots, A_n$ is the recognition framework in which the elements are mutually exclusive. The basic probability assignment (BPA) function is represented the m function

$$m : 2^{\Theta} \to [0, 1] \tag{4}$$

in which the m function satisfies

$$\begin{cases} m(\phi) = 0 \\ \sum_{A\in\Theta} m(A) = 1 \end{cases} \tag{5}$$

For $\forall A \subseteq \Theta$, the fusing rule of two functions m_1, m_2 in Θ is

$$m_1 \oplus m_2(A) = \frac{1}{K}\sum_{A_1\cap A_2 = A} m_1(A_1)\cdot m_2(A_2) \tag{6}$$

where K is the normalized constant

$$K = \sum_{A_1\cap A_2\neq\varnothing} m_1(A_1)\cdot m_2(A_2) = 1 - \sum_{A_1\cap A_2=\varnothing} m_1(A_1)\cdot m_2(A_2) \tag{7}$$

Analogously, if there are n functions $m_1, m_2, \dots, m_n$ in Θ, the fusing rule will be

$$\left(m_1 \oplus m_2 \oplus \cdots \oplus m_n(A)\right) = \frac{1}{K}\sum_{A_1\cap A_2\cap\ldots\cap A_n = A} m_1(A_1)\cdot m_2(A_2)\cdot\ldots\cdot m_n(A_n) \tag{8}$$

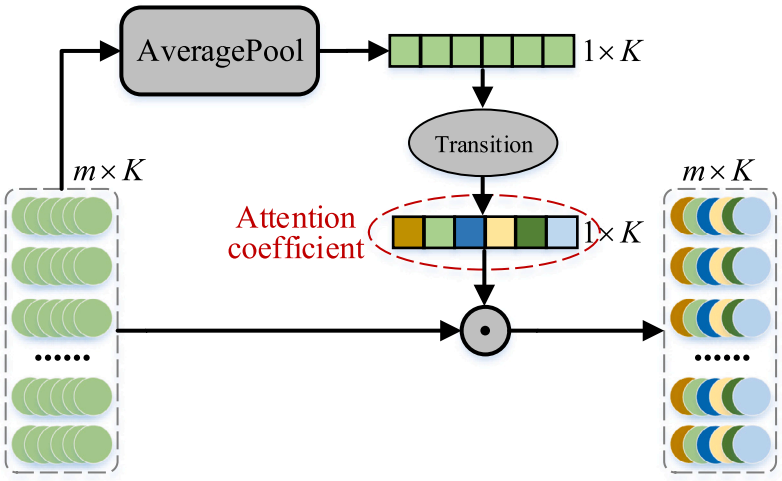


Fig. 1. Basic principle of attention mechanism.

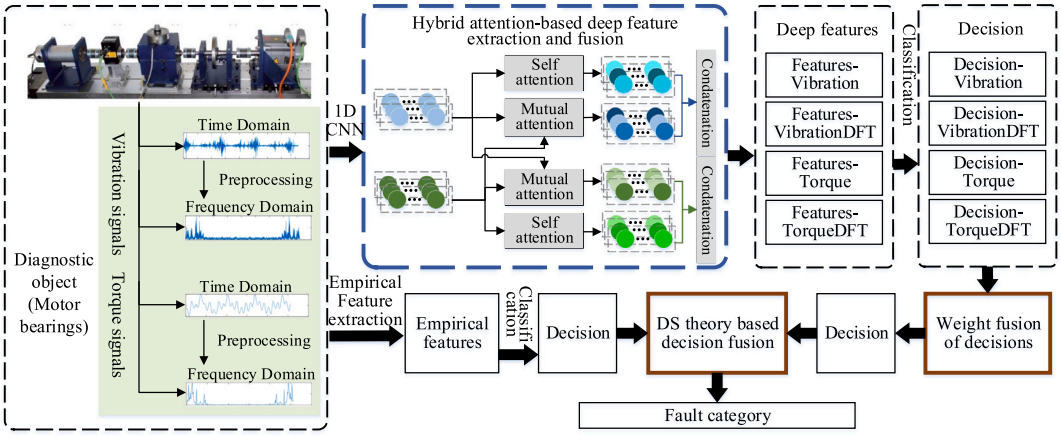


Fig. 2. The framework of MLIFN.

where K can be calculated as follows

$$K = \sum_{A_1 \cap A_2 \cap \ldots \cap A_n \neq \varnothing} m_1(A_1) \cdot m_2(A_2) \cdot \ldots \cdot m_n(A_n) = 1 - \sum_{A_1 \cap A_2 \cap \ldots \cap A_n = \varnothing} m_1(A_1) \cdot m_2(A_2) \cdot \ldots \cdot m_n(A_n) \tag{9}$$

3. Proposed method

The schematic of MLIFN is shown in Fig. 2. The MLIFN consists of four channel inputs which are two sensor signals (vibration signals and torque signals) in time domain and frequency domain. On the whole, the classification results of deep features and empirical features are fused based on standard deviation weighting strategy and DS theory. To be specific, a hybrid attention module composed of self attention and mutual attention is proposed to extract deep features of signal collected from multi-sensor, and the priori knowledge is applied to extract empirical features. In particular, the MLIFN can be divided into two processes. (1) Deep feature and empirical feature extraction, (2) fault classification and decision fusion.

3.1. Deep feature and empirical feature extraction

In this step, deep features are extracted from vibration and torque data in time and frequency domain. A hybrid attention based CNN which is composed of self attention, mutual attention and CNN is proposed to extract deep features, where the schematic of hybrid attention module is shown in Fig. 3.

In time domain, suppose that the vibration data and torque data in time domain are $X_1 \in \mathbb{R}^{M\times 1}$ and $X_2 \in \mathbb{R}^{N\times 1}$, respectively. At first, 1D CNN is applied to extract features as follows

$$y_k^1 = f\left(b_k^1 + \mathrm{cov}1D(W_k^1, X_1)\right), k = 1, 2, \ldots, K \tag{10}$$

$$y_k^2 = f\left(b_k^2 + \mathrm{cov}\,1D(W_k^2, X_2)\right), k = 1, 2, \ldots, K \tag{11}$$

where y_k^1 represents the feature extracted by the kth 1D convolution kernel of X_1, and K is the number of 1D convolution kernels. Then the maximum pooling and batch normalization were carried out to them

$$y_k^1 = BN(MP(y_k^1)), k = 1, 2, \ldots, K \tag{12}$$

$$y_k^2 = BN(MP(y_k^2)), k = 1, 2, \ldots, K \tag{13}$$

where $BN(\cdot)$ and $MP(\cdot)$ denote the batch normalization and maximum pooling, respectively. Repeat the above operations three times to obtain the fault features $F_{Vibration} = \left[f_{vib}^1, f_{vib}^2, \ldots, f_{vib}^K\right] \in R^{m\times K}$ and $F_{Torque} = \left[f_{tor}^1, f_{tor}^2, \ldots, f_{tor}^K\right] \in R^{m\times K}$.

Afterwards, the hybrid attention module is used to implement feature weighting. Firstly, F_{Torque} is transposed for weighting the features

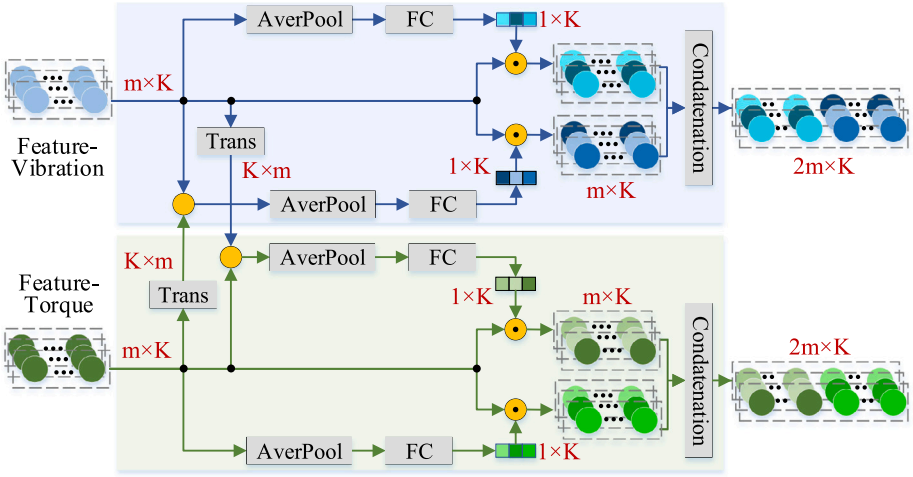


Fig. 3. Basic principle of attention mechanism.

of $F_{Vibration}$

$$F_{Torque} = Trans(F_{Torque}) \in R^{K\times m} \tag{14}$$

where Trans(∗) is transposition, which converts a row vector to a column vector. Then multiply F_{Torque} by $F_{Vibration}$ as follows

$$F_{MulAtt} = F_{Torque} \times F_{Vibration} \in R^{K\times K} \tag{15}$$

Matrix multiplication can realize the fine-grained fusion of features. The attention coefficient is to ensure that the generated attention coefficient fully considers the features of another signal. Then an average pooling and two full connection layers are employed to convert tensor F_{MulAtt} to the attention coefficient

$$F_{MulAtt} = Averpool(F_{MulAtt}) \in R^{1\times K} \tag{16}$$

$$A_{MulAtt} = FC\left(FC\left(F_{MulAtt}\right) \in R^{1\times K/r}\right) \in R^{1\times K} \tag{17}$$

where FC(∗) and r represent full connection and dimension reduction ratio, respectively, and the $A_{MulAtt} = [a_1, a_2, \ldots, a_K]$ is coefficient of mutual attention. Then the features of $F_{Vibration}$ are weighted by A_{MulAtt}

$$F_{Vibration}^{MulAtt} = [f_{vib}^1 a_1, f_{vib}^2 a_2, \ldots, f_{vib}^K a_K] \in R^{m\times K} \tag{18}$$

Secondly, a self attention is applied to weight the features of $F_{Vibration}$.

$$F_{SelAtt} = Averpool(F_{Vibration}) \in R^{1\times K} \tag{19}$$

$$A_{SelAtt} = FC\left(FC\left(F_{SelAtt}\right) \in R^{1\times K/r}\right) \in R^{1\times K} \tag{20}$$

where the $A_{SelAtt} = [a'_1, a'_2, \ldots, a'_K]$ is coefficient of self attention. Then the features of $F_{Vibration}$ are weighted by A_{SelAtt}

$$F_{Vibration}^{SefAtt} = [f_{vib}^1 a'_1, f_{vib}^2 a'_2, \ldots, f_{vib}^K a'_K] \in R^{m\times K} \tag{21}$$

Finally, $F_{Vibration}^{MulAtt}$ and $F_{Vibration}^{SefAtt}$ are concatenated

$$F_{Vibration}^{HybAtt} = Concat[(F_{Vibration}^{MulAtt}, F_{Vibration}^{SefAtt}), \text{axis} = 1] \in R^{2m\times K} \tag{22}$$

Where the $Concat(\cdot)$ is concatenation, and the parameter "axis" is the dimension to be concatenated. By replacing $F_{Vibration}$ and F_{Torque} in formulas (14)–(22) with each other, the feature F_{torque}^{HybAtt} will be obtained.

Denote the $X_3 \in \mathbb{R}^{M\times 1}$ and $X_4 \in \mathbb{R}^{N\times 1}$ as the vibration and torque data in frequency domain. Similarly, the higher-level features $F_{Vibration'} = \left[f_{vib'}^1, f_{vib'}^2, \ldots, f_{vib'}^K\right] \in R^{m\times K}$ and $F_{Torque'} = \left[f_{tor'}^1, f_{tor'}^2, \ldots, f_{tor'}^K\right] \in R^{m\times K}$ are extracted by formulas (10)–(13), where "$Vibration'$" represents vibration data in frequency domain. Afterwards, the hybrid attention module is applied to weight the features $F_{Vibration'}$ and $F_{Torque'}$ to obtain features $F_{Vibration'}^{HybAtt} \in \mathbb{R}^{2M\times K}$ and $F_{Torque'}^{HybAtt} \in \mathbb{R}^{2M\times K}$ through formulas (14)–(22).

Table 1
Empirical feature set.

Name	Equation	Name	Equation
Square mean root	$p_1 = \left(\frac{1}{N}\sum\sqrt{\lvert x\rvert}\right)^2$	Mean-absolute	$p_2 = \frac{1}{N}\sum\lvert x\rvert$
Root mean square	$p_3 = \sqrt{\frac{1}{N}\sum x^2}$	Kurtosis	$p_4 = \frac{\frac{1}{N}\sum x^4}{p_3{}^4}$
Skewness	$p_5 = \frac{\frac{1}{N}\sum x^3}{p_3^3}$	Crest factor	$p_6 = \frac{\max(\lvert x\rvert)}{p_3}$
Shape factor	$p_7 = \frac{p_3}{p_2}$	Clearance factor	$p_8 = \frac{\max(\lvert x\rvert)}{p_1}$
Impulse indicator	$p_9 = \frac{\max(\lvert x\rvert)}{p_2}$		

Statistical methods can provide handcraft features to traditional machine learning methods, which can transform complex signals into quantifiable indicators, and provide efficient tools for fault diagnosis. For example, in time domain, the mean can reflect the average energy level of the signal, and the variance represents the degree of signal fluctuation. In frequency domain, high variance may indicate abnormal energy concentration in a certain frequency band, and kurtosis can help determine whether there is a sharp abnormal frequency component. However, it is difficult for a single statistic to fully describe complex faults, so it is necessary to design empirical features that include multiple statistics. Reference [25] proves the validation of following empirical feature set as shown in Table 1.

The empirical features are extracted from vibration and torque data in time and frequency domain, which are denoted as $F_{Vibration}^{\text{EmpFea}} \in R^{9\times 1}$, $F_{VibrationDFT}^{\text{EmpFea}} \in R^{9\times 1}$, $F_{Torque}^{\text{EmpFea}} \in R^{9\times 1}$, $F_{TorqueDFT}^{\text{EmpFea}} \in R^{9\times 1}$. Then all the empirical features are concatenated as follows

$$F_{EnpFea} = Concat[(F_{Vibration}^{EmpFea}, F_{VibrationDFT}^{EmpFea}, F_{Torque}^{EmpFea}, F_{TorqueDFT}^{EmpFea}), \text{axis} = 0] \in \mathbb{R}^{36\times 1} \tag{23}$$

where F is the final empirical features.

3.2. Fault classification and decision fusion

Softmax regression is applied to fault classification of deep features and empirical features, in which the features are mapped to probabilities corresponding to bearing patterns. Assume that $O = [o_1, o_2, \ldots, o_N]$ is the original output of the last neural layer in which N is the number of bearing pattern, the probability can be calculated by softmax

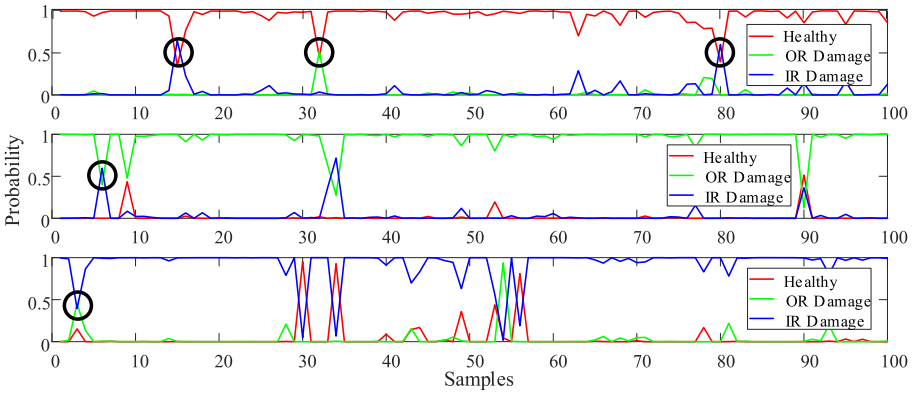


Fig. 4. The output probability of softmax regression.

function

$$p_j = \frac{e^{o_j}}{\sum_{i=1}^{N} e^{o_i}}, j = 1, 2, \ldots, N \tag{24}$$

where p_j is the probability of jth bearing pattern.

Taking three categories for example, we believe that under the premise of correct classification, the output probability of (1,0,0) gives a greater contribution to decision fusion than that of (0.55, 0.45, 0), and the standard deviation of (1,0,0) is much higher than that of (0.55, 0.45, 0). In addition, when the classifier is relatively ideal, the standard deviation of a part of misclassified samples is very small, as the sample of the black circle shown in Fig. 4, in which 300 samples is employed to obtain the output probability of softmax regression. The three subgraphs from above to below correspond to three bearing patterns of healthy, OR damage, and IR damage. Therefore, a weighted fusion of decision strategy is applied to increase the weight of decision with large standard deviations for better fusion effect.

Assume that the classification results of four deep features are $D_{Vibration}^{HybAtt}$, $D_{VibrationDFT}^{HybAtt}$, D_{Torque}^{HybAtt}, $D_{TorqueDFT}^{HybAtt}$ and that of empirical feature is D_{EmpFea}. Firstly, a standard deviation weighting strategy is employed to fuse the classification results of deep features. The standard deviation of four classification results of deep features are calculated as follows

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (p_i - \mu)^2}{N}} \tag{25}$$

Where μ is the mean of all the elements. Four standard deviations $\sigma_{Vibration}^{HybAtt}$, $\sigma_{VibrationDFT}^{HybAtt}$, σ_{Torque}^{HybAtt}, $\sigma_{TorqueDFT}^{HybAtt}$ are obtained. Then they are applied to weight and sum the four fault classification of deep features

$$D_{DeepFea} = \sigma_1 D_{Vibration}^{HybAtt} + \sigma_2 D_{VibrationDFT}^{HybAtt} + \sigma_3 D_{Torque}^{HybAtt} + \sigma_4 D_{TorqueDFT}^{HybAtt} \tag{26}$$

where $\sigma_1 = \frac{\sigma_{Vibration}^{HybAtt}}{\sigma_{all}}$, $\sigma_2 = \frac{\sigma_{VibrationDFT}^{HybAtt}}{\sigma_{all}}$, $\sigma_3 = \frac{\sigma_{Torque}^{HybAtt}}{\sigma_{all}}$, $\sigma_4 = \frac{\sigma_{TorqueDFT}^{HybAtt}}{\sigma_{all}}$ and $\sigma_{all} = \sigma_{Vibration}^{HybAtt} + \sigma_{VibrationDFT}^{HybAtt} + \sigma_{Torque}^{HybAtt} + \sigma_{TorqueDFT}^{HybAtt}$.

Afterwards, DS theory is employed to fuse the D_{EmpFea} and $D_{DeepFea}$

$$D_{out} = D_{EmpFea\alpha} \oplus D_{DeepFea\alpha}(c_i) = \frac{D_{EmpFea\alpha}(c_i) \cdot D_{DeepFea\alpha}(c_i)}{\sum_{i=1}^{N} D_{EmpFea\alpha}(c_i) \cdot D_{DeepFea\alpha}(c_i)}, \quad i = 1, 2, \ldots, N \tag{27}$$

where c_i represents the ith bearing pattern. The D_{out} is the final output of MLIFN.

4. Experimental validation

4.1. Data set description

To validate the effectiveness of MLIFN, the Paderborn bearing fault dataset introduced in [26] is used for experimentation. This dataset

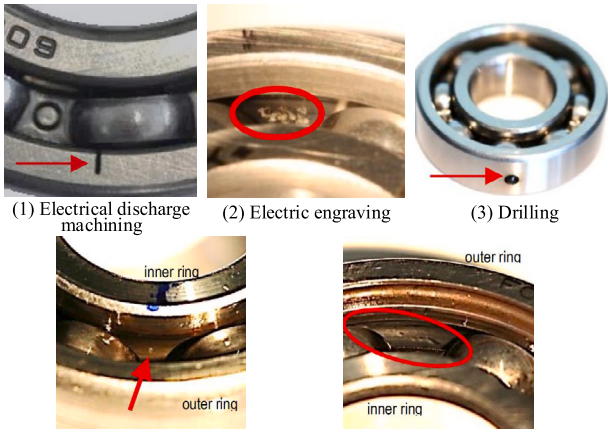


Fig. 5. Different damages of Bearing.

contains both artificial and real damages on ball bearing 6203. The artificial damages are generated through electric discharge machining, manual electric engraving, and drilling, as shown in 1–3 of Fig. 5. The real damages are caused by an accelerated lifetime test rig, as shown in 4 and 5 of Fig. 5. The outer ring (OR) and inner ring (IR) damages are caused by an accelerated lifetime test rig. Subsequently, those bearings are mounted on a modular test rig, as depicted in Fig. 6.

Research has shown that bearing fault will have the following consequences: (1) the introduction of a particular radial rotor movement and (2) load torque variations caused by the bearing fault [27]. The two consequences will cause a change in the vibration and torque signals, respectively. However, when different faults occur, the vibration signal and torque signal change differently. For instance, when nonuniform damage (such as crack and dent) occurs, the vibration signal will have obvious changes, which are shown as periodic variation of amplitude. And the size of the variation can reflect the severity of the fault. But the torque signal will not change significantly. By contrast, when the uniform damage (such as uniform wear) occurs, the rotating resistance of the bearing increases, resulting in a significant change in the torque signal. Although the amplitude of the vibration signal increased, the change rule of waveform is same as that of the healthy bearing. Consequently, vibration and torque signals exhibit complementary features aiming at different faults of bearing. Therefore, vibration and torque signals are utilized for bearing fault diagnosis.

In the Paderborn dataset, vibration signals are recorded at the adapter located at the top end of the rolling bearing module using a piezoelectric accelerometer (model no. 336C04, PCB Piezotronics, Inc.) with a sampling rate of 64 kHz. Simultaneously, torque signals

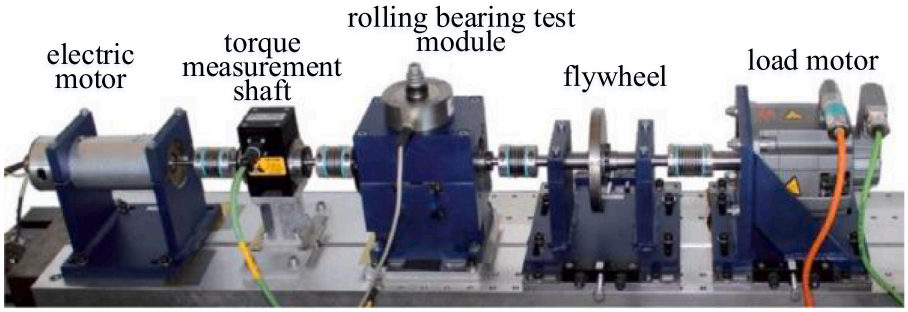


Fig. 6. Modular test rig.

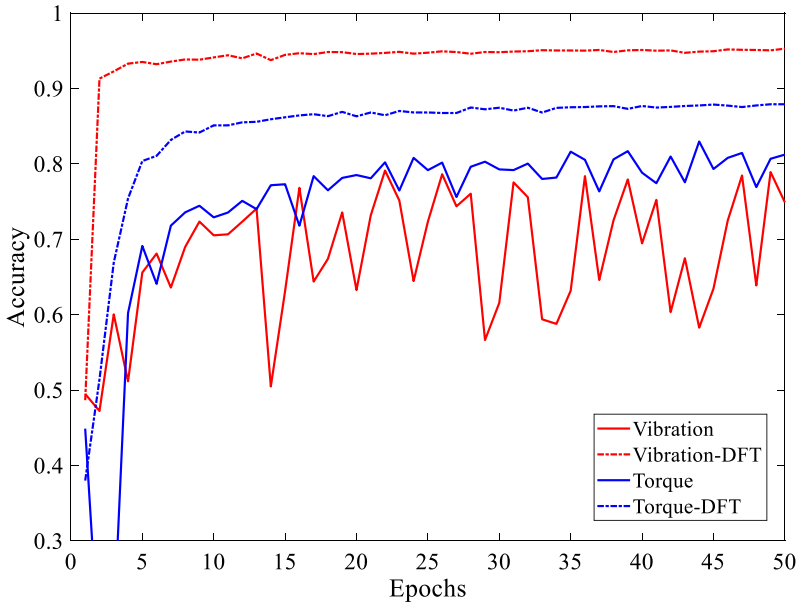


Fig. 7. Experimental results of vibration and torque signals.

are acquired using a torque transducer (Torque Transducer model 305, Magtrol, with a capacity of 2 Nm) at a sampling frequency of 4 kHz. In order to guarantee the dimension of feature is controllable when convolution pooling module (the length of pooling window is set to 2 or 4) is used for feature extraction, the length of the segmented data block is set to a multiple of 2. The experiments are conducted with different length of the segmented data block, including 512, 1024, 2048, 4096 and 8192. The experimental results show that the diagnostic performance of 4096 and 8192 is significantly better than the other three, but the computational complexity of 8192 is much higher than 4096. Therefore, 4096 is applied for vibration signal segments, while the torque signal segments are set to length of 256 due to their different sampling frequencies.

4.2. Experimental design and results analysis

4.2.1. Ablation experiments

A CNN with three convolution-pooling modules is employed to evaluate the diagnostic performance of single signal data. The convolutional kernels within these modules are set to (16, 7), (32, 5), and (64, 3), respectively. Here, (16, 7) indicates 16 convolutional kernels with a length of 7. The purpose of these kernels is to apply convolution operations on the input to extract fault features. Fivefold cross-validation is used in a single experiment to yield the results. In each experiment, 80% of the data is used for training and 20% for testing. The classification results of the test data sets are displayed in Fig. 7, where "Epochs" represent the number of times all data is reused. "Vibration-DFT" indicates that the input data is a vibration signal processed by discrete Fourier transform (DFT). Four identical inputs are fed into the same network to obtain the experimental results presented in Fig. 7. It is evident that whether it is the vibration signal or the torque signal, the experimental results in the time domain exhibit significant fluctuations, whereas the experimental results in the frequency domain demonstrate a marked increase in stability and accuracy. But we suspect that the torque data and vibration data contain different fault information. To fully leverage this, we utilize the MLIFN to deeply mine and fuse the different fault features extracted from the two signals, enhancing fault diagnosis accuracy.

The MLIFN incorporates a hybrid attention to facilitate the interaction between the feature extraction processes of vibration and torque data. The diagnostic performance of the hybrid attention is demonstrated in Fig. 8, revealing an increase in diagnostic accuracy but a decrease in stability of torque data. However, when compared to the original implementation, both the diagnostic accuracy and stability of vibration data are significantly improved after applying hybrid attention. Besides, the hybrid attention, self attention and mutual attention are respectively applied to fuse the deep features extracted from vibration and torque data for fault diagnosis. The comparison results are shown in Fig. 9. The diagnostic performance of self attention is much better than that of mutual attention, while combining self attention and mutual attention, the diagnostic performance of hybrid attention is significantly improved.

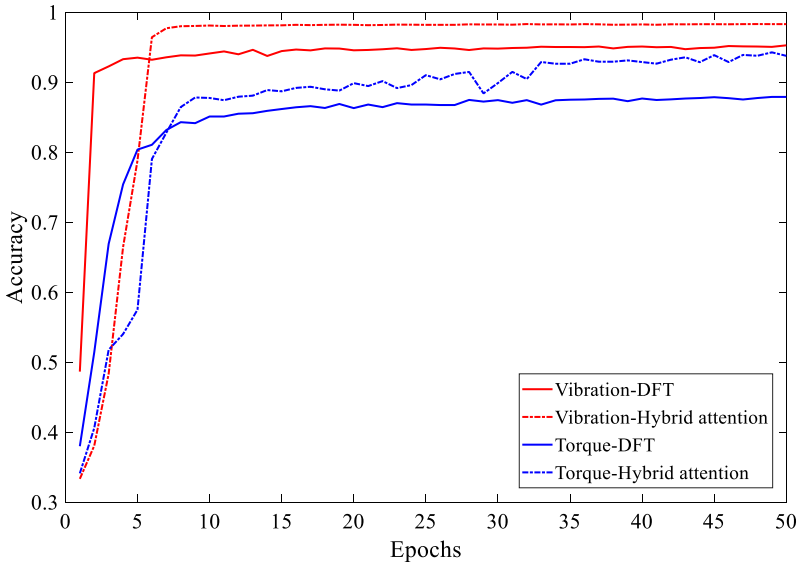


Fig. 8. Diagnostic accuracy of deep features extracted by hybrid attention.

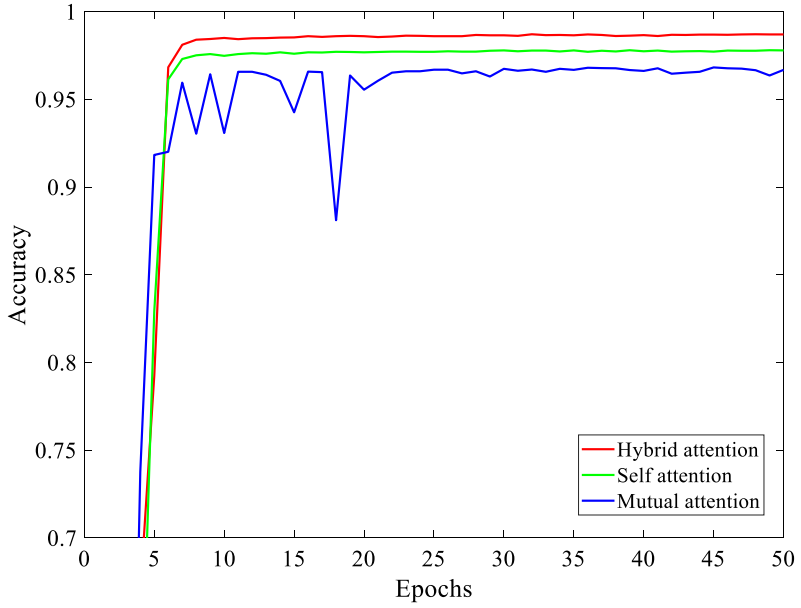


Fig. 9. Comparison of hybrid attention with self attention and mutual attention.

There are two level decision fusion in MLIFN, which are fused decision of deep features by weight fusion, and fused decision of deep features and empirical features. Two experiments are conducted to evaluate the performance of the two decision fusion as shown in Figs. 10 and 11. It can be seen from Fig. 10 that both average fusion and weight fusion can improve the diagnostic accuracy, but the accuracy and training stability of weighted fusion are higher. And Fig. 11 shows that after fusing deep features and empirical features, the diagnostic performance is significantly improved, especially the vibration signal.

Fig. 12 shows the results of ablation experiments of MLIFN. It can be seen that both mutual attention mechanisms and weight fusion of decisions can improve the performance of fault diagnosis. After applying the hybrid attention, the diagnostic accuracy of vibration and torque signals has been improved by 3.03% and 5.78%, respectively. When the hybrid attention is used for deep feature extraction and the classification results are fused by weight fusion, the diagnostic accuracy reaches 99.12%. Combining all the innovations mentioned above, the accuracy of the MLIFN reaches 99.82%. Fig. 13 shows the accuracy curve and loss curve of MLIFN. It proves the convergence of the MLIFN, where the MLIFN has already converged before 20 epochs of training. Besides, in order to validate the real-time performance of MLIFN, the validation experiments for timeliness and computational complexity are conducted, and the results are shown in Table 2. Although the computational complexity of MLIFN is increased, the timeliness can still be guaranteed.

The data set from Paderborn contains various conditions, including the rotational speed of the drive system, the radial force on the test bearing, and the load torque in the drive train. These parameters are listed in Table 3. To validate the ability to generalize of MLIFN, the data set is divided into training and testing sets based on different working conditions. Each experiment is conducted five times. The results of the experiments are shown in Table 4, where the "0–1" condition means that data from condition 0 are used for training and data from condition 1 are used for testing. Seven out of the twelve cross-domain experiments had results higher than 90%, with "2–0" and "0–2" having results higher than 99%. In terms of the Paderborn data set, the change

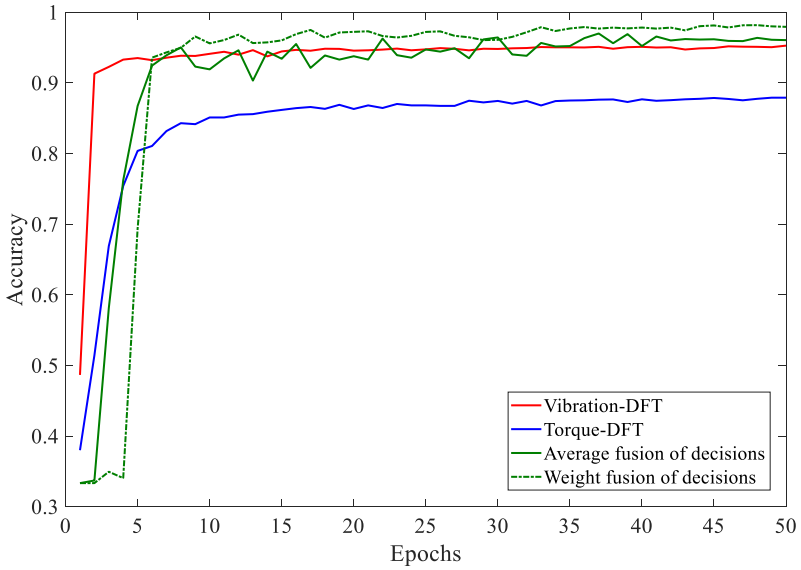


Fig. 10. Diagnostic accuracy of fused decisions of deep features.

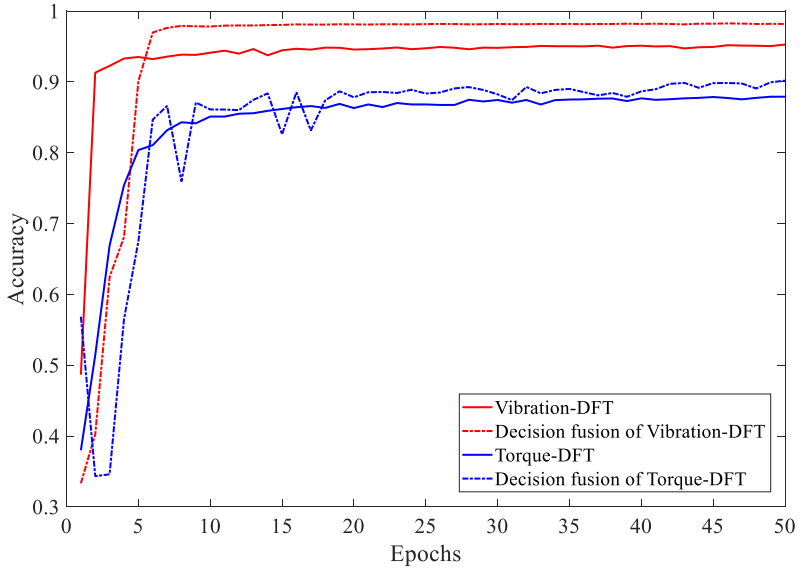


Fig. 11. Diagnostic accuracy of fused decisions of deep features and empirical features.

Table 2
Diagnostic accuracy of deep features extracted by hybrid attention.

Method	Number of parameters	Total time of training
Vibration-DFT	38,563	888 s
Torque-DFT	16,675	391 s
Vibration-Hybrid attention	236,163	891 s
Torque-Hybrid attention	236,163	909 s
Hybrid attention & Weight fusion of decisions	434,828	975 s
MLIFN	434,939	1015 s

Table 3
Paderborn data set under different working conditions.

No.	Rotational speed [rpm]	Load torque [Nm]	Radial force [N]	Name of Setting
0	1500	0.7	1000	N15_M07_F10
1	900	0.7	1000	N09_M07_F10
2	1500	0.1	1000	N15_M01_F10
3	1500	0.7	400	N15_M07_F04

in rotational speed has the greatest impact on the data, and the change in load torque has the least impact. In terms of the method, MLIFN has a strong ability to generalize.

4.2.2. Comparison experiments

The conducted experiments investigate the performance of traditional machine learning techniques applied to vibration signals from the Paderborn dataset. Features in the time domain (e.g., root mean square, crest factor), frequency domain (e.g., peak amplitude, FFT energy), and time–frequency domain (e.g., wavelet packet decomposition energy, detail coefficients one and two) were extracted and combined for fault classification. Comparative analyses were performed on six fusion methods: data-level fusion [4,28], feature-level fusion [20,29], and decision-level fusion [21,30]. The results, based on the means and standard deviations of fivefold cross-validation, are summarized in Table 5. It is evident that traditional ML methods achieve a maximum diagnostic accuracy of 95.66%, albeit with high standard deviation, indicating

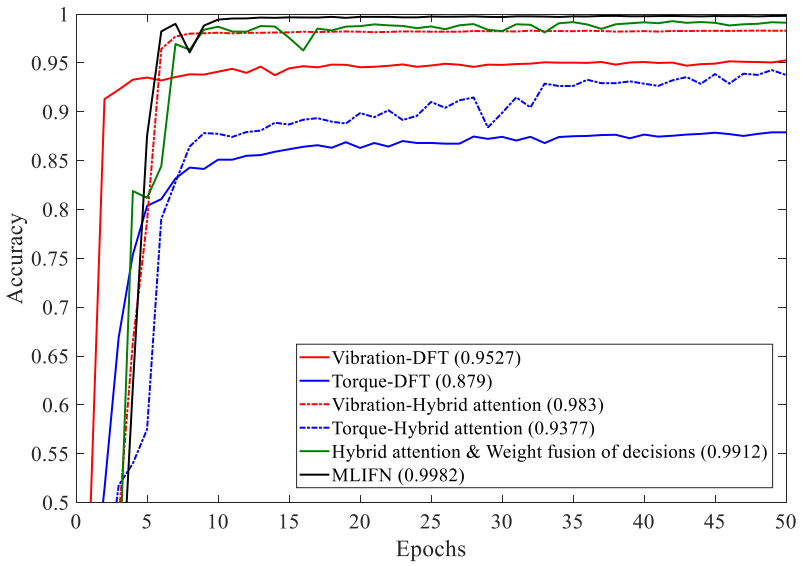


Fig. 12. Results of ablation experiments of MLIFN.

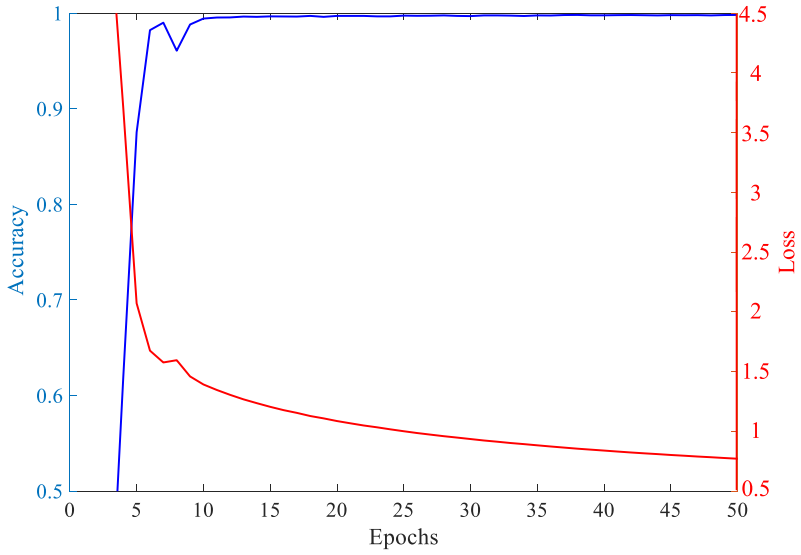


Fig. 13. The accuracy curve and loss curve of MLIFN.

Table 4
Experiment results under different working conditions.

Working condition	Accuracy (%)	Working condition	Accuracy (%)	Working condition	Accuracy (%)
0–1	68.75 ±1.87	0–2	99.78 ±0.16	0–3	90.09 ±0.63
1–0	69.63 ±1.40	1–2	70.95 ±0.84	1–3	90.13 ±1.19
2–0	99.61 ±0.25	2–1	75.55 ±0.92	2–3	90.58 ±0.55
3–0	91.53 ±0.93	3–1	88.18 ±1.23	3–2	92.34 ±0.85

variability in performance. Data fusion [4,28] exhibit limitations due to the loss of the original data structure, while feature fusion [20,29] suffer from the dependency on extractor design. Although decision-level fusion [21] achieves accuracy of over 98% and [30] achieves accuracy of almost 99%, they remain inferior to the proposed MLIFN, which effectively leverages hybrid attention and two-level decision fusion to exploit complementary fault information from vibration and torque signals.

Given the constraints of accelerated lifetime test rigs, replicating all potential fault conditions using real damaged bearings is challenging. To address this, subsequent experiments utilized data from artificially and real damaged bearings for training and testing, with results presented in Table 6. Traditional ML methods exhibited low accuracy and high standard deviation due to the lack of representativeness in real damaged bearing features. Besides, the performance of the data fusion [4,28] and feature fusion [29] methods is also unsatisfactory. The performance of feature fusion [20] and decision fusion [21,30] methods have been improved, but is still not ideal. By contrast, MLIFN achieved a diagnostic accuracy of 90.02% with minimal standard deviation, demonstrating its robustness and superior fault diagnostic capability under varying conditions.

Table 5
Classification accuracy of different methods.

Algorithm		Accuracy (%)
Traditional ML methods	CART	95.66 ± 1.57
	RF	96.07 ± 1.16
	NN	50.68 ± 5.88
	ELM	57.76 ± 4.12
	KNN	59.95 ± 2.66
Data fusion method in [4]		96.83 ± 0.76
Data fusion method in [28]		97.02 ± 1.28
Feature fusion method in [20]		97.39 ± 0.97
Feature fusion method in [29]		97.26 ± 0.69
Decision fusion method in [21]		98.12 ± 0.39
Decision fusion method in [30]		98.96 ± 0.29
MLIFN		99.82 ± 0.07

Table 6
Classification accuracy of different methods with artificial damaged bearing data for training and real damaged bearing data for testing.

Algorithm		Accuracy (%)
Traditional ML methods	CART	65.79 ± 3.98
	RF	63.28 ± 4.41
	NN	66.12 ± 5.34
	ELM	64.28 ± 8.12
	KNN	63.36 ± 3.69
Data fusion method in [4]		76.59 ± 2.58
Data fusion method in [28]		73.69 ± 4.36
Feature fusion method in [20]		83.69 ± 2.24
Feature fusion method in [29]		79.98 ± 3.36
Decision fusion method in [21]		85.02 ± 1.98
Decision fusion method in [30]		83.26 ± 1.38
MLIFN		90.02 ± 1.05

5. Conclusion

This paper presents a multi-level information fusion network (MLIFN) for bearing fault diagnosis. In the network, the empirical features and deep features are classified and then fused by a two level decision fusion. Besides, a hybrid attention which consists of self attention and mutual attention is applied to mine correlation among the features extracted from multi-sensor data through the interaction of feature extraction. The Paderborn bearing dataset is employed to validate the effectiveness of the MLIFN. Results show that the diagnostic accuracy of MLIFN is much better than the other methods which can be up to 99.82%.

In the future, we will study the data fusion methods under unbalanced sample conditions. Different decision fusion methods will be emphatically studied to improve the diagnostic performance under unbalanced sample conditions.

CRediT authorship contribution statement

Daichao Wang: Writing – original draft, Methodology, Formal analysis. **Yan Song:** Formal analysis. **Jinhao Xing:** Software. **Yinghao Zhuang:** Software. **Jingbo Zhao:** Methodology. **Yibin Li:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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